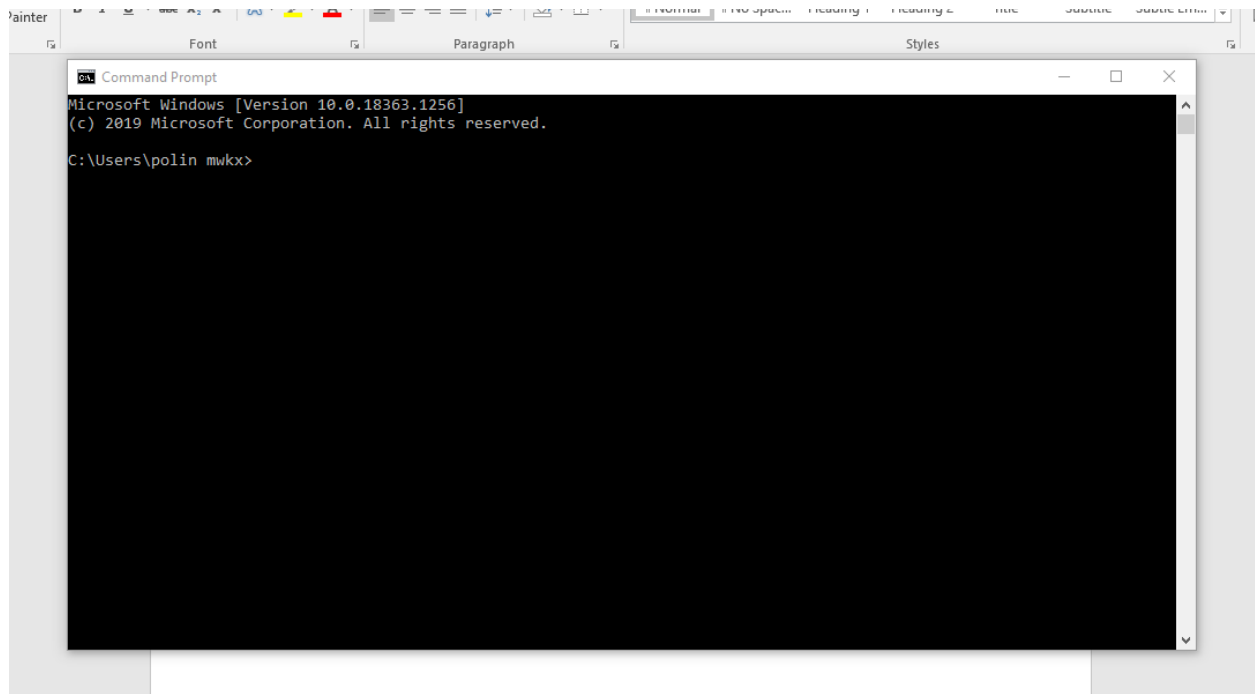


NB. ENSURE YOUR XAMPP SERVER IS OPEN

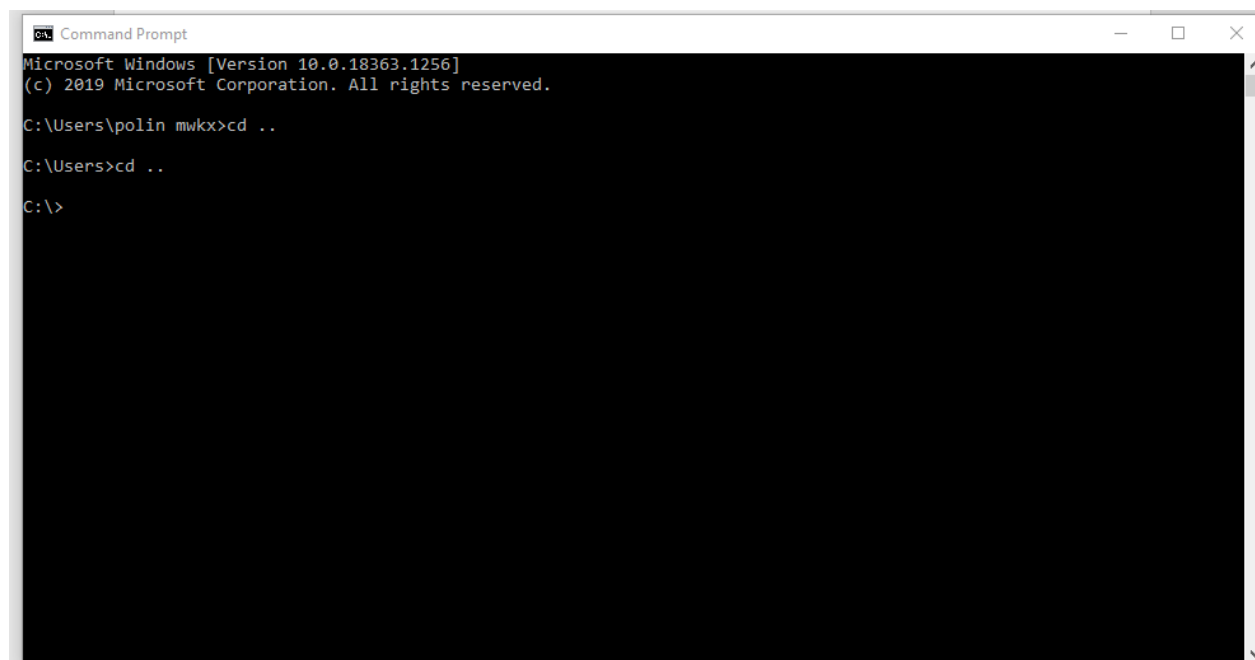
Open command prompt.



Then type `cd ..` command (it returns you to a prior directory) and press enter

Type again `cd ..` and press enter

This should take you to C drive directory



Then navigate to where your local host by typing the following commands

cd xampp\mysql\bin then press enter

```
C:\ Command Prompt
Microsoft Windows [Version 10.0.18363.1256]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\polin mwkx>cd ..

C:\Users>cd ..

C:\>cd xampp\mysql\bin

C:\xampp\mysql\bin>
```

Then type mysql -u root -p then enter the password (by default if you have not manipulated your xampp, it has No password so simply press enter, but if you changed the password then enter the password and press enter.)

```
C:\ Command Prompt - mysql -u root -p
Microsoft Windows [Version 10.0.18363.1256]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\polin mwkx>cd ..

C:\Users>cd ..

C:\>cd xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 22
Server version: 10.4.14-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

SHOW DATABASES PRESENT IN YOUR XAMPP

The show command helps us to display all the databases resent in our databases

Syntax: SHOW DATABASES;

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| phpmyadmin |
| test |
+-----+
5 rows in set (0.035 sec)
```

CREATE A NEW DATABASE

Syntax: CREATE DATABASE DATABASE_NAME;

Lets create a database called Student_database

Then use show command to check whether the database was created

```
MariaDB [(none)]> CREATE DATABASE Student_database;
Query OK, 1 row affected (0.056 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| phpmyadmin |
| student_database |
| test |
+-----+
6 rows in set (0.001 sec)

MariaDB [(none)]>
```

We have to select which database we will use for our table creation. To do this we use the command USE

Syntax: USE DATABASE_NAME

```
MariaDB [(none)]> USE Student_database
Database changed
MariaDB [Student_database]>
```

Now under in this database we want to create a table called student_records

ID	NAME	CLASS	MARK	SEX
1	John Deo	Four	75	female
2	Max Ruin	Three	85	male
3	Arnold	Three	55	male
4	Krish Star	Four	60	female
5	John Mike	Four	60	female
6	Alex John	Four	55	male
7	My John Rob	Fifth	78	male
8	Asruid	Five	85	male
9	Tes Qry	Six	78	male
10	Big John	Four	55	female

From the data given we can assign the following data types and constraints:

ID -> INT(10) datatype and PRIMARY KEY constraint

NAME-> Varchar(50) and NOT NULL constraint

Class-> Varchar (10) and NOT NULL constraint

Mark ->INT(10) and NOT NULL CONTRAINT

SEX -> Varchar(20) and NOT NULL constraint

Lets us create the table

The syntax for creating a table is

```
CREATE TABLE table_name (
column1 datatype constraint,
column2 datatype constraint,
column3 datatype constraint,
.....
columnN datatype constraint,
PRIMARY KEY (column_name)
);
```

```
MariaDB [student_database]> CREATE TABLE student_records(ID INT(10) UNIQUE, NAME VARCHAR(50) NOT NULL, CLASS VARCHAR(50) NOT NULL, MARK INT(10) NOT NULL, SEX VARCHAR(20) NOT NULL, PRIMARY KEY (ID));
Query OK, 0 rows affected (0.296 sec)
```

Having created the table successfully, we can now see its structure by using a command DESC

Syntax: DESC table_name;

```
MariaDB [student_database]> Desc student_records;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int(10)       | NO   | PRI | NULL    |       |
| NAME  | varchar(50)   | NO   |     | NULL    |       |
| CLASS | varchar(50)   | NO   |     | NULL    |       |
| MARK  | int(10)       | NO   |     | NULL    |       |
| SEX   | varchar(20)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.085 sec)

MariaDB [student_database]>
```

Now we can insert data into the table we created

Lets insert the first row of the table

ID	NAME	CLASS	MARK	SEX
1	John Deo	Four	75	female

Insert command syntax:

INSERT INTO table_name (column1, column2....columnN) VALUES (value1, value2....valueN);

```
MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (1,'John Doe','Four',75,'Female')
;
Query OK, 1 row affected (0.061 sec)

MariaDB [student_database]>
```

NB. The strings are usually quoted using single quotes eg 'john doe'

We have inserted the first data let's check whether the data has been stored in our database: we use the select command to select all the columns and data in it.

Syntax: SELECT * FROM table_name

```

MariaDB [student_database]> SELECT * FROM student_records;
+----+-----+-----+-----+-----+
| ID | NAME   | CLASS | MARK | SEX   |
+----+-----+-----+-----+-----+
| 1  | John Doe | Four  | 75   | Female |
+----+-----+-----+-----+-----+
1 row in set (0.039 sec)

MariaDB [student_database]>

```

Lets add the second row of data

ID	NAME	CLASS	MARK	SEX
1	John Deo	Four	75	female
2	Max Ruin	Three	85	male

```

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (2,'Max Ruin','Three',85,'Male');
Query OK, 1 row affected (0.057 sec)

MariaDB [student_database]>

```

Lets insert the other rows

```

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (3,'Arnold','Three',55,'Male');
Query OK, 1 row affected (0.071 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (4,'Krish Star','Four',60,'Female');
Query OK, 1 row affected (0.041 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (5,'John Mike','Four',60,'Female');
Query OK, 1 row affected (0.064 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (6,'Alex John','Four',55,'Male');
Query OK, 1 row affected (0.072 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (7,'My John Rob','Fifth',78,'Male');
Query OK, 1 row affected (0.065 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (8,'Asruid','Five',85,'Male');
Query OK, 1 row affected (0.059 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (9,'Tes Qry','Six',78,'Male');
Query OK, 1 row affected (0.068 sec)

MariaDB [student_database]> INSERT INTO student_records(ID,NAME,CLASS,MARK,SEX) VALUES (10,'Big John','Four',55,'Female');
Query OK, 1 row affected (0.099 sec)

MariaDB [student_database]>

```

Lets see how the table looks like using the select command

We are going to select and display all the data in our table

SELECT*FROM student_records;

```
MariaDB [student_database]> SELECT * FROM student_records;
```

ID	NAME	CLASS	MARK	SEX
1	John Doe	Four	75	Female
2	Max Ruin	Three	85	Male
3	Arnold	Three	55	Male
4	Krish Star	Four	60	Female
5	John Mike	Four	60	Female
6	Alex John	Four	55	Male
7	My John Rob	Fifth	78	Male
8	Asruid	Five	85	Male
9	Tes Qry	Six	78	Male
10	Big John	Four	55	Female

10 rows in set (0.001 sec)

Select only name and mark columns

SELECT NAME, MARK FROM students_records;

```
MariaDB [student_database]> SELECT NAME, MARK FROM student_records;
```

NAME	MARK
John Doe	75
Max Ruin	85
Arnold	55
Krish Star	60
John Mike	60
Alex John	55
My John Rob	78
Asruid	85
Tes Qry	78
Big John	55

10 rows in set (0.001 sec)

```
MariaDB [student_database]>
```

Select only the class column

SELECT CLASS FROM student_records;

```

MariaDB [student_database]> SELECT CLASS FROM student_records;
+-----+
| CLASS |
+-----+
| Four  |
| Three |
| Three |
| Four  |
| Four  |
| Four  |
| Fifth |
| Five  |
| Six   |
| Four  |
+-----+
10 rows in set (0.001 sec)

MariaDB [student_database]>

```

Select the records of students who are in class four

SELECT *FROM student_records WHERE CLASS='Four';

```

MariaDB [student_database]> SELECT *FROM student_records WHERE CLASS='Four';
+-----+-----+-----+-----+-----+
| ID | NAME      | CLASS | MARK | SEX   |
+-----+-----+-----+-----+-----+
| 1  | John Doe  | Four  | 75   | Female |
| 4  | Krish Star | Four  | 60   | Female |
| 5  | John Mike | Four  | 60   | Female |
| 6  | Alex John | Four  | 55   | Male   |
| 10 | Big John  | Four  | 55   | Female |
+-----+-----+-----+-----+-----+
5 rows in set (0.055 sec)

MariaDB [student_database]>

```

Select records of students who belong to class Three and mark is 85

SELECT *FROM student_records WHERE CLASS='Three' AND MARK=85;

```

MariaDB [student_database]> SELECT *FROM student_records WHERE CLASS='Three' AND MARK=85;
+-----+-----+-----+-----+-----+
| ID | NAME      | CLASS | MARK | SEX   |
+-----+-----+-----+-----+-----+
| 2  | Max Ruin  | Three | 85   | Male  |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [student_database]>

```

Lets select the records of all female students

```
SELECT *FROM student_records WHERE SEX='Female';
```

```
MariaDB [student_database]> SELECT *FROM student_records WHERE SEX='Female';
+-----+-----+-----+-----+-----+
| ID | NAME      | CLASS | MARK | SEX   |
+-----+-----+-----+-----+-----+
| 1  | John Doe  | Four  | 75   | Female|
| 4  | Krish Star| Four  | 60   | Female|
| 5  | John Mike | Four  | 60   | Female|
| 10 | Big John  | Four  | 55   | Female|
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [student_database]>
```

Alter command

It can be used to change the name of the table

Lets change the name of our table to pupils_records

Syntax

```
ALTER TABLE table_name RENAME TO new_table_name;
```

```
ALTER TABLE student_records RENAME TO pupils_records;
```

Then afterwards type SHOW tables; to check whether the table has been renamed.

```
MariaDB [student_database]> ALTER TABLE student_records RENAME TO pupils_records;
Query OK, 0 rows affected (0.406 sec)

MariaDB [student_database]> show tables;
+-----+
| Tables_in_student_database |
+-----+
| pupils_records             |
+-----+
1 row in set (0.001 sec)

MariaDB [student_database]>
```

Can also be used to add, drop,or modify a column

Syntax

```
ALTER TABLE table_name {ADD|DROP|MODIFY} column_name {data_type};
```

ALTER TABLE pupils_records ADD UNITS VARCHAR(10); (N.B after running this command , then check the table structure by running DESC pupils_records to check whether the column has been added)

```
MariaDB [student_database]> DESC pupils_records;
```

Field	Type	Null	Key	Default	Extra
ID	int(10)	NO	PRI	NULL	
NAME	varchar(50)	NO		NULL	
CLASS	varchar(50)	NO		NULL	
MARK	int(10)	NO		NULL	
SEX	varchar(20)	NO		NULL	
UNITS	varchar(10)	YES		NULL	

```
6 rows in set (0.085 sec)

MariaDB [student_database]>
```

From the output we can see that the null field of UNITS is yes meaning it can be null, we can modify the constraint to not null

```
ALTER TABLE pupils_records MODIFY UNITS VARCHAR(10) NOT NULL;
```

```
MariaDB [student_database]> ALTER TABLE pupils_records MODIFY UNITS VARCHAR(10) NOT NULL;
Query OK, 0 rows affected, 10 warnings (1.220 sec)
Records: 0 Duplicates: 0 Warnings: 10

MariaDB [student_database]> DESC pupils_records;
```

Field	Type	Null	Key	Default	Extra
ID	int(10)	NO	PRI	NULL	
NAME	varchar(50)	NO		NULL	
CLASS	varchar(50)	NO		NULL	
MARK	int(10)	NO		NULL	
SEX	varchar(20)	NO		NULL	
UNITS	varchar(10)	NO		NULL	

```
6 rows in set (0.047 sec)

MariaDB [student_database]>
```

We can also drop a column using the alter command

```
ALTER TABLE pupils_records DROP UNITS ;
```

```

MariaDB [student_database]> ALTER TABLE pupils_records DROP UNITS;
Query OK, 0 rows affected (0.104 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [student_database]> DESC pupils_records;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int(10)       | NO   | PRI | NULL    |       |
| NAME  | varchar(50)   | NO   |     | NULL    |       |
| CLASS | varchar(50)   | NO   |     | NULL    |       |
| MARK  | int(10)       | NO   |     | NULL    |       |
| SEX   | varchar(20)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.050 sec)

MariaDB [student_database]>

```

UPDATE COMMAND

Used to update data stored in a table

SYNTAX

UPDATE table_name SET column1 = value1, column2 = value2....columnN=valueN

[WHERE CONDITION];

ID	NAME	CLASS	MARK	SEX
1	John Deo	Four	75	female

Lets update the above column data to CLASS = Five, MARK = 80 and NAME= Richard Deo

UPDATE pupils_records SET NAME= "Richard Doe", CLASS= "Five", SEX="Male"

WHERE ID=1;

After running this command select all the data and display to check the update

```

MariaDB [student_database]> UPDATE pupils_records SET NAME= "Richard Doe", CLASS= "Five", SEX="Male"
-> WHERE ID=1;
Query OK, 1 row affected (0.143 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [student_database]> SELECT*FROM pupils_records;
+-----+-----+-----+-----+-----+
| ID | NAME      | CLASS | MARK | SEX  |
+-----+-----+-----+-----+-----+
| 1  | Richard Doe | Five  | 75   | Male |
| 2  | Max Ruin   | Three | 85   | Male |
| 3  | Arnold     | Three | 55   | Male |
| 4  | Krish Star | Four  | 60   | Female |
| 5  | John Mike  | Four  | 60   | Female |
| 6  | Alex John  | Four  | 55   | Male |
| 7  | My John Rob | Fifth | 78   | Male |
| 8  | Asruid     | Five  | 85   | Male |
| 9  | Tes Qry    | Six   | 78   | Male |
| 10 | Big John   | Four  | 55   | Female |
+-----+-----+-----+-----+-----+
10 rows in set (0.002 sec)

MariaDB [student_database]>

```

You can update other records in the table

DROP COMMAND AND TRUNCATE COMMAND

The difference between these two commands is that truncate command deleted the contents of a table but retains its structure while DROP command deletes the table contents together with its structure.

Run truncate command first then select and display all the data using SELECT COMMAND then check the table structure using the DESC command. Do the same for the delete command and check the difference

Syntax: Truncate table_name; DROP TABLE table_name;

```

MariaDB [student_database]> truncate pupils_records;
Query OK, 0 rows affected (2.143 sec)

MariaDB [student_database]> SELECT*FROM pupils_records;
Empty set (0.008 sec)

MariaDB [student_database]> DESC pupils_records;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int(10)   | NO   | PRI | NULL    |       |
| NAME  | varchar(50) | NO   |     | NULL    |       |
| CLASS | varchar(50) | NO   |     | NULL    |       |
| MARK  | int(10)   | NO   |     | NULL    |       |
| SEX   | varchar(20) | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (1.208 sec)

MariaDB [student_database]>

```

```
MariaDB [student_database]> DROP TABLE pupils_records;  
Query OK, 0 rows affected (0.189 sec)  
  
MariaDB [student_database]> show tables;  
Empty set (0.001 sec)  
  
MariaDB [student_database]> SELECT*FROM pupils_records;  
ERROR 1146 (42S02): Table 'student_database.pupils_records' doesn't exist  
MariaDB [student_database]>
```

You can also use DROP command to delete the database

DROP DATABASE database_name;